

Zihan Hei

CV

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EDUCATION

Binghamton University, State University of New York

Expected June 2027

Bachelor of Mathematical Sciences in Statistics; Minor in Digital and Data Studies

GPA: 3.8/4.0

RESEARCH INTERESTS

Biostatistics; artificial intelligence for health; bioinformatics; statistical learning; cancer biomarker discovery; mental health evaluation; mixed-methods research; reproducible data science; interactive data visualization.

WORK EXPERIENCE

Quantitative Intern, First-Year Research Immersion Program

June 2026–Present

Binghamton University

Faculty Mentors: Dr. Megan Fegley and Dr. Caitlin Light

- Contribute to the development of interactive R Shiny dashboards for tracking student outcomes and research dissemination metrics for program evaluation and reporting.
- Develop an integrated database to organize and track dissemination records across FRI research programs.
- Design and implement Qualtrics surveys to collect research dissemination outcomes, including publications and presentations.
- Develop Google Apps Script automation workflows to process survey responses, organize author information, and streamline reporting.

Mental Health Workshop Quantitative Intern

June 2026–Present

Binghamton University

Faculty Mentor: Dr. Shane McCarty

- Design Qualtrics surveys to evaluate mental health literacy, well-being, nonviolent communication, and workshop effectiveness.
- Develop quantitative evaluation methods and survey instruments for the Fall 2026 mental health workshop.
- Create interactive data visualizations and individualized wellness profiles using Observable JS.

RESEARCH EXPERIENCE

Chemo-Immunotherapy Biomarker Discovery

**November 2025–
Present**

Independent Study, Binghamton University

Faculty Mentor: Dr. Nancy Guo

- Reproduce and validate single-cell RNA-seq preprocessing pipelines, including quality control, normalization, and high-variance gene selection.
- Apply Non-negative Matrix Factorization to identify gene signatures and temporal expression patterns.
- Perform differential expression analysis and Gene Set Enrichment Analysis to interpret biological pathways associated with treatment response; manuscript in preparation.

Health Information Avoidance

August 2025–Present

Independent Study, Binghamton University

Faculty Mentors: Dr. Shane McCarty and Dr. Heather Orom

- Evaluate model performance using logistic regression, multivariate adaptive regression splines, and random forests to predict health-information avoidance using the tidymodels and caret packages in R.
- Develop a reproducible scientific report using Quarto and LaTeX; manuscript in preparation.

A Mixed-Methods Analysis of Promotion and Prevention Mindsets

**August 2025–
November 2025**

Undergraduate Research, Binghamton University

Faculty Mentor: Dr. Shane McCarty

- Applied logistic regression modeling in R to investigate whether promotion and prevention mindsets predict health- and safety-related goal preferences.

- Collaborated on a mixed-methods research project combining quantitative analyses with qualitative coding to examine public conceptions of community health and safety.
- Served as first author on a poster abstract accepted by the 2026 American Public Health Association Annual Meeting.

Zero-Sum Beliefs

June 2025–Present

Undergraduate Research, Binghamton University

Faculty Mentor: Dr. Shane McCarty

- Examine political party affiliation and zero-sum beliefs as predictors of voter choice in the 2024 U.S. election.
- Conduct ANOVA/Kruskal-Wallis tests, logistic regression, and machine-learning classifiers; manuscript in preparation.

First-Year Research Immersion: Community and Global Public Health

August 2023–
December 2024

Undergraduate Research, Binghamton University

Faculty Mentor: Dr. Shane McCarty

- Collaborated with a seven-person team to research the impact of endometriosis on women's mental health, culminating in two poster presentations to peers and faculty in December 2023 and 2024.
- Conducted quantitative data analysis using SPSS for six or more hours weekly, collaborating with peers to develop studies on vaccine hesitancy among pregnant women.
- Designed and distributed a comprehensive campus-wide survey using Qualtrics, collecting and analyzing responses.

PUBLICATIONS & TECHNICAL WRITING

- McCarty, S., Light, C., Hei, Z., & Fegley, M. (2026). FRI Peer Mentor Mindsets Guide. Binghamton University.
- McCarty, S., & Hei, Z. (2025). Selected chapters. In S. McCarty (Ed.), *The Quantitative Playbook for Public Health Research in R*.

PRESENTATIONS & CONFERENCE ABSTRACTS

- Hei, Z., Giallella, J., & McCarty, S. (2026). A Mixed-Methods Analysis of Promotion and Prevention Mindsets. Poster abstract accepted by the American Public Health Association Annual Meeting.
- McCarty, S., Hei, Z., Ruolo, G., Khalap, I., & Giallella, J. (2025). Promotion & prevention mindsets: A four-study, mixed-methods approach. Invited online presentation, Society for Community Research & Action.
- Public Knowledge & Attitudes on Vaccinations and Reproductive Health. FRI Poster Research Session, Binghamton University, 2024.
- Endometriosis and Women's Mental Health. FRI Poster Research Session, Binghamton University, 2023.

MENTORING

R Programming Peer Mentor

January 2025–May
2025

First-Year Research Immersion Program, Binghamton University

Faculty Mentor: Dr. Shane McCarty

- Guided students in applying foundational R programming skills toward team-based research projects.
- Collaborated with the instructor and other research assistants to support research teams' Quarto research reports.

TECHNICAL PROJECTS

Histopathology Thyroid Cancer Detection Using YOLO

January 2026–May
2026

Group Project, Binghamton University

- Built a YOLO-based deep-learning model to detect thyroid cancer, abnormal, and normal tissue in histopathological images for potential pre-screening applications.
- Evaluated model performance through Type I and Type II error analysis, assessing risks of missed diagnoses and unnecessary follow-up.
- Interpreted confusion matrices to assess model limitations related to class imbalance and dataset representativeness.

Deep Reinforcement Learning in Board Games

September 2025–
December 2025

Group Project, Binghamton University

- Implemented deep reinforcement-learning agents in PyTorch to play Tic-Tac-Toe and Connect Four through self-play.
- Designed neural networks mapping board states to action values and visualized training performance using Matplotlib.

HONORS & AWARDS

- NSF Research Experiences for Undergraduates Research Stipend Recipient, 2026–2027
- Pi Mu Epsilon, National Mathematics Honor Society, 2026
- Harpur College Dean's List, Every Semester Attended, Fall 2023–Present

TECHNICAL SKILLS

Programming: R (tidyverse), Python (Pandas, NumPy), Google Apps Script

Research Reproducibility: Quarto, GitHub, LaTeX

Data Visualization: ggplot2, Plotly, Matplotlib, Observable JS

Machine Learning: Random forests, decision trees, classification models, deep learning, scikit-learn, tidymodels, caret, PyTorch, YOLO

Bioinformatics: Differential expression analysis, limma, Gene Set Enrichment Analysis, RNA-seq workflows

Survey and Analysis Tools: Qualtrics, SPSS